

**INTERNSHIP PROJECT REPORT**

**Farmer-to-Consumer Marketplace Management System**

Submitted By

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## **CERTIFICATE**

This is to certify that Ms. **Rutuja Shelke** has successfully completed the internship project titled "**Farmer-to-Consumer Marketplace Management System**" as part of the Full Stack Development Internship. The project work was carried out sincerely and submitted in partial fulfillment of the internship requirements.

## **DECLARATION**

I hereby declare that the project report entitled "**Farmer-to-Consumer Marketplace Management System**" is my original work carried out during the internship period. The information presented in this report is true to the best of my knowledge and has not been submitted elsewhere for any academic award.

**Name:** Rutuja Shelke

## **ACKNOWLEDGEMENT**

I express my sincere gratitude to Upskill Campus for providing me with the opportunity to undertake this internship project. I thank all the mentors and coordinators who guided me throughout the internship period.

I also extend my thanks to my faculty members, friends, and family members for their support and encouragement during the development of this project.

The experience gained through this internship has enhanced my technical knowledge and practical understanding of software development.

## **ABSTRACT**

The Farmer-to-Consumer Marketplace Management System is a Core Java application designed to establish a direct connection between farmers and consumers. The system enables farmers to add products, manage inventory, monitor earnings, and update product details. Consumers can browse products, search for available items, purchase products, and view order history.

The project aims to reduce dependency on intermediaries and provide a transparent marketplace for agricultural products. The application is developed using Core Java concepts such as classes, objects, collections, loops, conditional statements, and menu-driven programming.

The system offers features including farmer management, product management, inventory control, order processing, earnings tracking, and product search functionality. The

implementation demonstrates the practical application of object-oriented programming concepts while addressing a real-world agricultural marketing problem.

Keywords: Core Java, Farmer Marketplace, Consumer System, Inventory Management, Order Processing, OOP.

## **TABLE OF CONTENTS**

1. Introduction
2. Objectives
3. Problem Statement
4. Existing System
5. Proposed System
6. Scope of Project
7. Software Requirements
8. Hardware Requirements
9. System Analysis

10. System Design
11. Architecture Diagram
12. Modules Description
13. Algorithm
14. Implementation
15. Testing
16. Outputs
17. Advantages
18. Limitations
19. Future Scope
20. Conclusion
21. References

## **CHAPTER 1: INTRODUCTION**

Agriculture plays a crucial role in the economy. Farmers often face challenges in selling their products due to the involvement of multiple intermediaries. These intermediaries reduce the profits earned by farmers and increase the prices paid by consumers.

The Farmer-to-Consumer Marketplace Management System is developed to provide a direct platform where farmers can showcase their products and consumers can purchase them directly. This approach ensures fair pricing and improves transparency in the supply chain.

The project is implemented using Core Java and demonstrates the use of object-oriented programming principles to create a practical and efficient marketplace management solution.

## **CHAPTER 2: OBJECTIVES**

The objectives of the project are:

- To provide a direct marketplace for farmers and consumers.
- To manage product inventories efficiently.
- To maintain order records.
- To track farmer earnings.
- To reduce dependency on middlemen.



- To implement a menu-driven system using Core Java.
- To demonstrate practical use of object-oriented programming concepts.

### **CHAPTER 3: PROBLEM STATEMENT**

Traditional agricultural marketing systems involve multiple intermediaries between farmers and consumers. This results in:

- Reduced farmer profits.
- Increased product prices.
- Lack of transparency.
- Poor inventory management.

- Limited market reach for farmers.

A software solution is required to simplify product management and provide a direct interaction platform.

## **CHAPTER 4: EXISTING SYSTEM**

The existing system relies heavily on local markets and intermediaries.

### **Disadvantages**

- Multiple middlemen.
- Higher consumer prices.
- Lower farmer earnings.
- Limited transparency.

- Inefficient inventory management.

## **CHAPTER 5: PROPOSED SYSTEM**

The proposed system provides:

- Farmer registration.
- Product listing.
- Product search.
- Product purchasing.
- Inventory updates.
- Earnings tracking.
- Order history management.

The application directly connects farmers and consumers through a centralized platform.

## **CHAPTER 6: SCOPE OF PROJECT**

The project can be used by:

- Farmers
- Consumers
- Agricultural cooperatives
- Small-scale marketplaces

The system can be extended into a web-based or mobile application in the future.

## **CHAPTER 7: SOFTWARE REQUIREMENTS**

### **Operating System**

- Windows 10/11

### **Programming Language**

- Java

### **IDE**

- VS Code

### **JDK**

- JDK 8 or Above

#### **Version Control**

- GitHub

## **CHAPTER 8: HARDWARE REQUIREMENTS**

### **Processor**

- Intel i3 or higher

### **RAM**

- 4 GB minimum

### **Storage**

- 20 GB free space

### **Display**

- Standard Monitor

## **CHAPTER 9: SYSTEM ANALYSIS**

The system uses a menu-driven approach to manage marketplace operations.

### **Inputs**

- Farmer Details
- Product Details
- Purchase Requests

### **Processing**

- Product Search
- Order Generation
- Inventory Update

- Earnings Calculation

### **Outputs**

- Product Lists
- Order Details
- Earnings Reports

## **CHAPTER 10: SYSTEM DESIGN**

### **Main Classes**

#### **Farmer**

Attributes:

- Farmer ID
- Farmer Name
- Earnings

#### **Product**

Attributes:

- Product ID
- Product Name
- Price
- Quantity

## Order

Attributes:

- Order ID
- Product Name
- Quantity
- Total Amount

## CHAPTER 11: ARCHITECTURE DIAGRAM

Farmer

|

v

Marketplace System

|

+----- Product Management

|

+----- Inventory Management

|



+----- Order Processing

|

+----- Earnings Dashboard

|

Consumer

## **CHAPTER 12: MODULE DESCRIPTION**

### **Farmer Module**

Functions:

- Add Farmer
- View Farmers

### **Product Module**

Functions:

- Add Product
- View Product
- Delete Product

## **Inventory Module**

Functions:

- Update Stock
- Check Availability

## **Order Module**

Functions:

- Buy Product
- Generate Order

## **Earnings Module**

Functions:

- Calculate Revenue
- Display Farmer Earnings

## **CHAPTER 13: ALGORITHM**

### **Add Farmer**

1. Accept farmer details.
2. Create Farmer object.
3. Store in ArrayList.
4. Display success message.

### **Add Product**

1. Select farmer.
2. Enter product details.
3. Store product information.

4. Confirm addition.

### **Buy Product**

1. Select product.
2. Enter quantity.
3. Verify stock.
4. Generate order.
5. Update inventory.
6. Update earnings.

## **CHAPTER 14: IMPLEMENTATION**

The application is implemented using:

- Classes
- Objects
- ArrayList
- Scanner
- Loops
- Conditional Statements
- Methods

The system follows object-oriented programming principles and provides modular functionality.

## **CHAPTER 15: TESTING**

### **Test Case 1**

Input:

Add Farmer

Expected Result:

Farmer Added Successfully

Status:

Pass

### **Test Case 2**

Input:

Add Product

Expected Result:

Product Added Successfully

Status:

Pass

### **Test Case 3**

Input:

Buy Product

Expected Result:

Order Generated

Status:

Pass

### **Test Case 4**

Input:

Update Inventory

Expected Result:

Inventory Updated

Status:

Pass

## **CHAPTER 16: OUTPUTS**

### **Output 1**

Main Menu Display

**Output 2**

Farmer Registration

**Output 3**

Product Listing

**Output 4**

Product Search

**Output 5**

Purchase Confirmation

**Output 6**

Order History

**Output 7**

Farmer Earnings Dashboard

(Insert screenshots here)

**CHAPTER 17: ADVANTAGES**

- Easy to use.
- Reduces intermediary involvement.
- Improves farmer earnings.
- Maintains inventory records.
- Provides order management.
- Demonstrates OOP concepts.

## **CHAPTER 18: LIMITATIONS**

- Console-based application.
- No database integration.
- No online payment system.
- No real-time notifications.

## **CHAPTER 19: FUTURE SCOPE**

- Web Application using Spring Boot.
- Mobile Application.
- Database Integration.
- Online Payment Gateway.
- Weather Integration.
- AI-Based Crop Recommendation.
- Real-Time Inventory Tracking.



## **CHAPTER 20: CONCLUSION**

The Farmer-to-Consumer Marketplace Management System successfully demonstrates the implementation of a marketplace platform using Core Java. The project enables farmers to manage products and earnings while allowing consumers to search and purchase products efficiently.

The application provides practical exposure to object-oriented programming concepts and showcases the use of Java collections and menu-driven application development. The project has significant potential for future enhancement and real-world implementation.

## **REFERENCES**

1. Herbert Schildt, Java: The Complete Reference.
2. Oracle Java Documentation.
3. Java Collections Framework Documentation.
4. Object-Oriented Programming Concepts.
5. Upskill Campus Internship Learning Materials.

## **GITHUB LINKS**

**Source Code Link**

<https://github.com/rutujashelke2109/upskillcampus/blob/main/FarmerConsumerMarketplace.java>

**Report Link**